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Secure Distribution Venture Pvt. Ltd.

What Is BOT / HAM Project Funding ?

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Executive Summary & Course Orientation

BOT (Build-Operate-Transfer) and HAM (Hybrid Annuity Model) Project Funding represent the bedrock of public-private partnership (PPP) infrastructure financing in India. Designed specifically to finance the development, civil construction, long-term operation, and maintenance of national highways, expressways, state corridors, metro rail extensions, urban bypasses, and municipal sewage treatment plants (STPs), these structured project finance facilities bridge the fundamental gap between heavy upfront capital expenditure (CAPEX) and multi-decade public utility operations.

This white paper document is structured as an authoritative masterclass reference manual. It provides an exhaustive, end-to-end framework covering every operational, legal, credit, underwriting, tax, and structural aspect of BOT and HAM Project Funding in India. Designed to serve as a complete standalone curriculum, reading this document alone will equip any finance professional, credit analyst, investment banker, corporate promoter, or student with the exact institutional knowledge required to master the product, structure complex project finance transactions, perform quantitative underwriting, and clear all professional certification assessments on infrastructure concession debt.

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Module 1: Product Overview & Core Fundamentals

1.1 Definition and Foundational Concept of BOT / HAM Funding

BOT (Build-Operate-Transfer) and HAM (Hybrid Annuity Model) Project Funding refers to long-term, limited-recourse or non-recourse debt facilities extended to a Special Purpose Vehicle (SPV) established specifically to execute a public infrastructure concession awarded by a government authority—such as the National Highways Authority of India (NHAI), Ministry of Road Transport and Highways (MoRTH), State Road Development Corporations (e.g., MSRDC, UPSHA, KRDCL), or Municipal Corporations under National Missions (e.g., Jal Ganga, AMRUT 2.0).

Unlike traditional corporate loans where creditworthiness is assessed on the parent promoter's balance sheet and historical P&L statements, BOT / HAM Project Funding is underwritten primarily on the standalone cash-generating capacity of the concession asset itself. The debt service is supported by contractually guaranteed cash inflows from the government concession authority (in the case of HAM and BOT-Annuity) or user fee toll collections (in the case of BOT-Toll), ring-fenced through a bank-controlled Trust and Retention Account (TRA) Escrow mechanism.

1.2 The Evolution of Highway & Public Infrastructure Concessions in India

To understand the dominance of the Hybrid Annuity Model (HAM) in Indian infrastructure debt markets today, one must trace the historical evolution of public infrastructure contracting frameworks across four distinct phases:

1. **Engineering, Procurement, and Construction (EPC) Model**: Under pure EPC, the government finances 100% of the project cost directly from the public exchequer. The private contractor is paid on a item-rate or lump-sum basis as civil milestones are completed. While EPC eliminates financial risk for the contractor, it imposes an overwhelming fiscal burden on government budgets and lacks long-term private sector accountability for asset quality and post-construction maintenance.
2. **Build-Operate-Transfer (BOT) Toll Model**: Introduced to attract private capital, BOT-Toll requires the private concessionaire to finance, construct, operate, and maintain the highway for a concession period (typically 20 to 30 years). In return, the concessionaire earns the right to levy and collect user tolls from vehicular traffic. While highly successful in the early 2000s, BOT-Toll stalled significantly between 2012 and 2015 due to aggressive traffic over-estimations, macro-economic slowdowns, land acquisition (Right of Way) delays, local toll resistance, and severe debt defaults that burdened Indian public sector banks with Non-Performing Assets (NPAs).

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3. **Build-Operate-Transfer (BOT) Annuity Model**: To eliminate vehicular traffic volume risk for developers, the BOT-Annuity model was created. Here, the private developer constructs the project and receives fixed, pre-determined bi-annual payments (annuities) from the government authority over 15 to 20 years, regardless of actual traffic volume. However, BOT-Annuity required the developer to arrange 100% of the upfront project capital (equity + debt) during construction, leading to severe liquidity bottlenecks for mid-sized contractors.
4. **Hybrid Annuity Model (HAM) - The 2016 Game-Changer**: Formulated by MoRTH and NHAI in 2016 under the leadership of the Union Government, HAM represents a hybrid synthesis of EPC and BOT-Annuity. Under HAM, the government authority provides **40% of the Bid Project Cost (BPC)** as a construction grant disbursed in five equal tranches of 8% tied to physical progress. The remaining **60% of the project cost** is arranged by the developer SPV through a combination of equity (typically 15% of total project cost) and senior bank debt (typically 45% of total project cost). Post-completion, the government pays the balance 60% of asset cost along with interest and inflation-indexed O&M fees through **30 semi-annual annuity payments over 15 years**.

1.3 The Standalone Special Purpose Vehicle (SPV) Architecture

A non-negotiable structural requirement of BOT / HAM Project Funding is the creation of a standalone Special Purpose Vehicle (SPV). The SPV is a legally independent, bankruptcy-remote corporate entity incorporated under the Companies Act, 2013.

The primary objectives of the SPV structure include: - **Bankruptcy Remoteness**: Isolates the infrastructure project from financial distress, insolvency, or liabilities originating from the parent promoter group or sister subsidiaries. - **Single-Asset Capital Structure**: Ensures that all assets, concessions, permits, bank accounts, and liabilities belong exclusively to the project entity, preventing co-mingling of funds. - **Contractual Ring-Fencing**: All project contracts—the Concession Agreement with NHAI, the Lump-Sum Turnkey (LSTK) EPC Contract, the O&M Agreement, the Substitution Agreement, and the TRA Escrow Agreement—are executed directly in the name of the SPV. - **Targeted Financial Leverage**: Allows high debt-to-equity ratios (up to 80:20 on the SPV outlay; effectively 70:30 on total project cost) that would be unacceptable on a corporate balance sheet.

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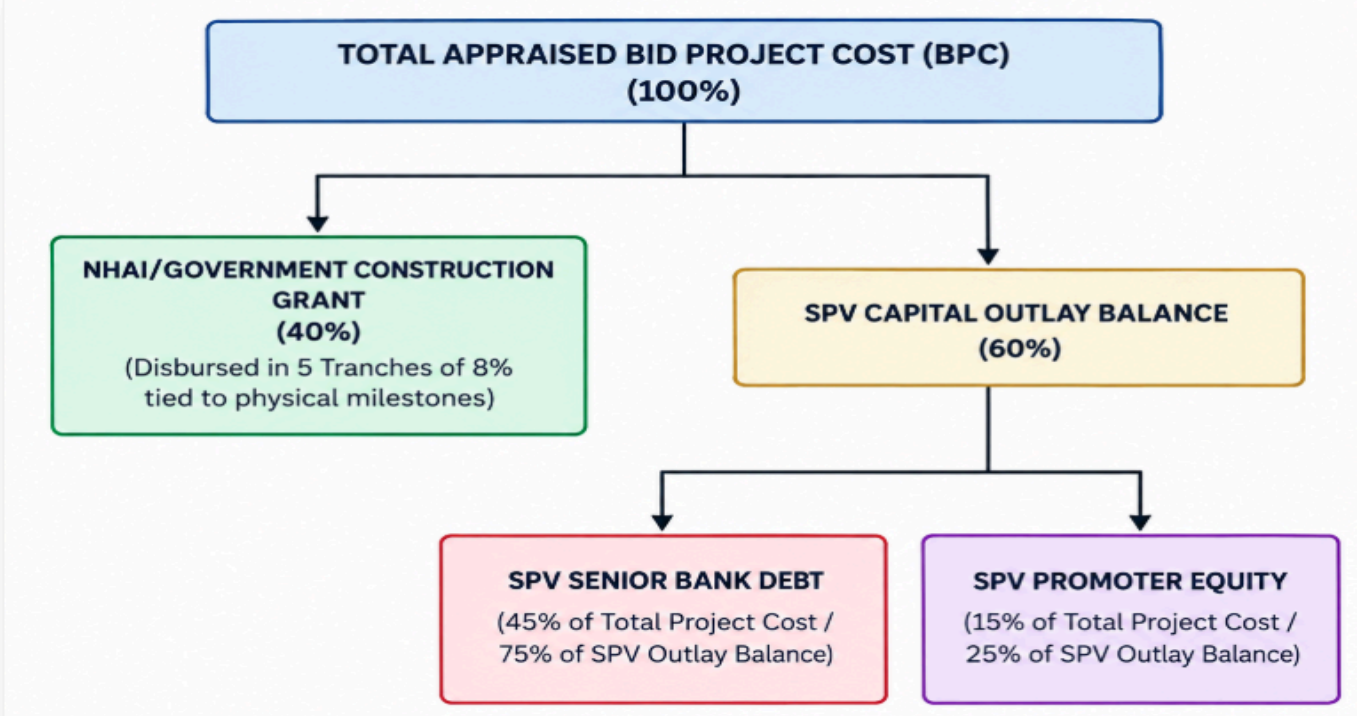
Module 2: Structural Architecture & How BOT / HAM Funding Works

2.1 The 40:60 Capital Split & Funding Flow in HAM Projects

The financial architecture of a HAM highway project is structured around a clear division of capital contribution between the Concession Authority (NHAI/MoRTH) and the

Developer SPV during the construction phase:

Detailed Breakdown of Capital Components:



1. **NHAI Construction Grant (40% of BPC):** Disbursed directly into the SPV's Escrow Account in five equal payments of 8% each, upon achieving 20%, 40%, 60%, 80%, and 100% physical construction progress as certified by the Lender's Independent Engineer (LIE) and NHAI Independent Engineer (IE).
2. **Developer Equity Contribution (15% of BPC / 25% of SPV Outlay)**: Sourced entirely from the promoter group. Lensders mandate that 50% to 100% of the promoter cash equity be injected upfront prior to the initial debt drawdown, or disbursed pro-rata alongside bank debt.
3. **Senior SPV Bank Debt (45% of BPC / 75% of SPV Outlay):** Provided by a consortium of commercial banks (e.g., SBI, Bank of Baroda, Punjab National Bank, Canara Bank) or specialized infrastructure lenders (e.g., NaBFID, REC,

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PFC). The debt is drawn down in milestone-based tranches to fund civil construction, land development, and interest during construction (IDC).

2.2 Operational Annuity Phase Mechanics & The Three Annuity Streams

Upon achieving the Commercial Operation Date (COD) or Provisional COD (PCOD), the project transitions from the high-risk construction phase to the low-risk operational annuity phase. Over a 15-year operational concession period (30 semi-annual cycles), NHAI remits **three distinct annuity components** directly into the SPV's Escrow Account:

$$extTotalBi - AnnualNHAIAnnuityReceipt = extAssetCostAnnuity + extInterestAnnuity + extInflation - IndexedO\&MAn$$

1. Asset Cost Annuity (40% Balance of Bid Project Cost)

NHAI pays back the remaining 60% balance of the Bid Project Cost (the portion funded by SPV debt and equity) in 30 equal semi-annual installments over 15 years.

$$extBi - AnnualAssetCostAnnuityInstallment = racextBidProjectCostimes0.6030$$

2. Interest Annuity on Reducing Balance

NHAI pays interest on the reducing balance of the unpaid 60% asset cost. To protect the developer against interest rate volatility, the interest rate paid by NHAI is contractually benchmarked:

$$extInterestAnnuityRate = extRBIBankRate + 3.00\% \quad ext(orAverage1 - YearMCLRoftop5ScheduledCommercialBanks)$$

This benchmark alignment ensures that whenever the RBI raises or lowers benchmark interest rates, NHAI's interest annuity payment automatically adjusts, preserving the bank's debt servicing spread.

3. Operation & Maintenance (O&M) Annuity

NHAI pays a dedicated bi-annual O&M fee based on the quoted O&M cost in the financial bid. To cushion against wage and material inflation, this O&M annuity is dynamically indexed to a composite inflation index:

$$extO\&MInflationIndex = (0.70imesextWPI) + (0.30imesextCPI)$$

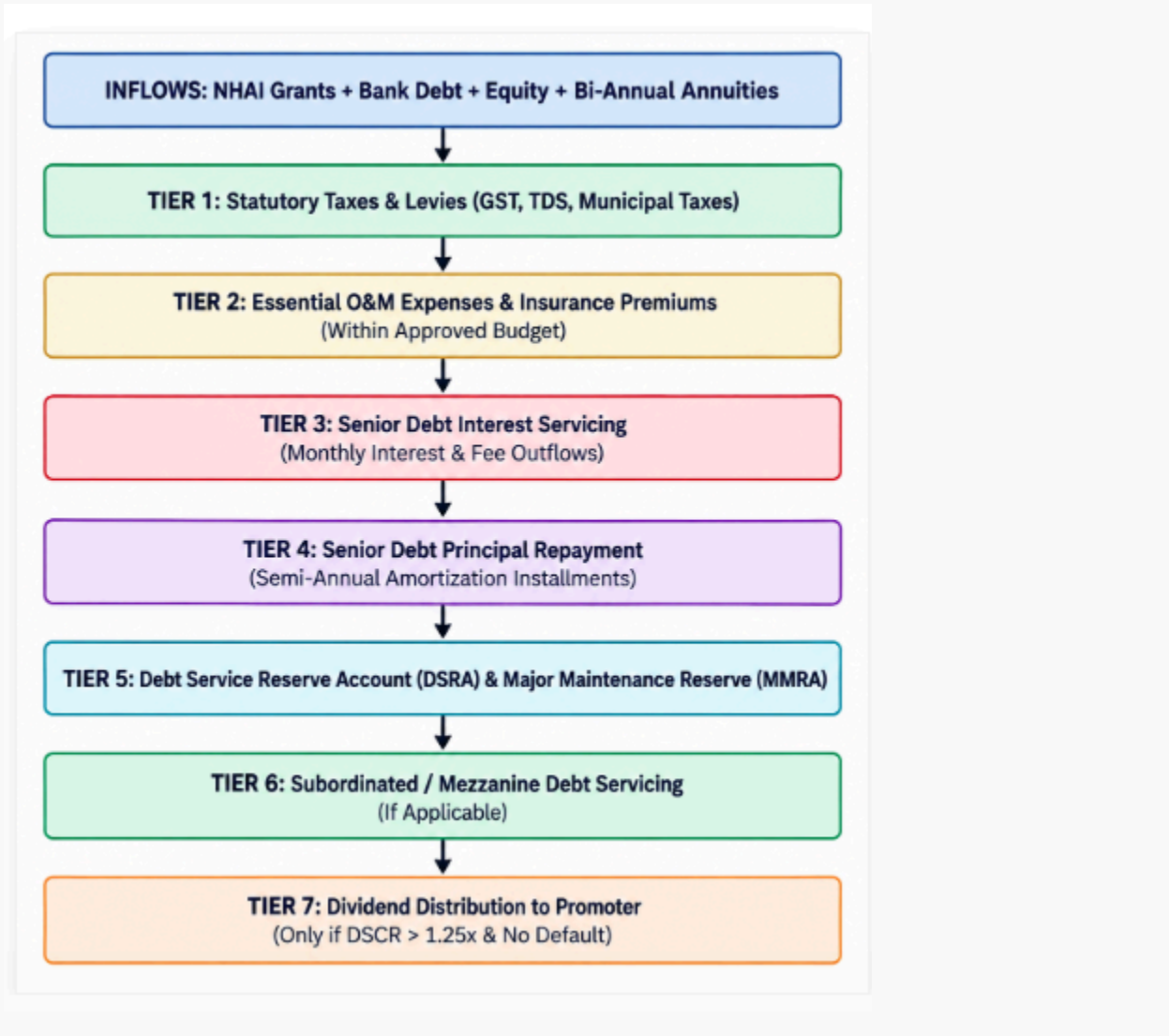
Where WPI is the Wholesale Price Index and CPI is the Consumer Price Index published by the Reserve Bank of India.

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2.3 The 7-Tier Trust and Retention Account (TRA) Escrow Waterfall

To guarantee absolute debt service priority, all project cash inflows—NHAI construction grants, bank debt drawdowns, promoter equity, NHAI bi-annual annuities, and interest income—must be deposited directly into a bank-controlled Trust and Retention Account (TRA) Escrow held with the Lead Lender.

The TRA agreement enforces a strict, legally binding 7-tier payment waterfall. Funds can only be released to lower tiers once higher tiers are 100% satisfied:



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Module 3: Comparative Capital Architecture Matrix

To evaluate how BOT / HAM Project Funding compares against other corporate credit and infrastructure financing structures, the matrix below details the key structural, financial, risk, and pricing parameters across Indian concession debt products:

Structural Parameter	HAM (Hybrid Annuity Model)	BOT-Toll Model	BOT-Annuity Model	Pure EPC Works Credit	Corporate Balance Sheet Term Loan	InvIT Post-COD Refinancing
Primary Borrower Entity	Standalone Project SPV	Standalone Project SPV	Standalone Project SPV	Parent EPC Contracting Co.	Parent Holding / Operating Co.	Infrastructure Investment Trust
Traffic / Volumetric Risk	**Zero Traffic Risk** (100% borne by NHAI)	**100% Traffic Risk** borne by Concessionaire	**Zero Traffic Risk** (100% borne by NHAI)	Zero (Civil contractor only)	Operational Business Risk	Zero Traffic Risk (HAM assets) / Mixed
Grant Support	**40% Construction Grant** from NHAI	0% to 20% Viability Gap Funding (VGF)	0% Grant (100% SPV funded upfront)	100% Funded via Monthly Progress Bills	None	Capital Recycling Proceeds
Maximum Permissible Debt	45% of Total Cost (75% of SPV Outlay)	65% - 70% of Total Cost	70% - 80% of Total Cost	20% - 25% Working Capital Line	50% - 65% LTV on Fixed Assets	70% Debt-to-Asset Cap (SEBI rules)
Promoter Equity Requirement	15% of Total Cost (25% of SPV Outlay)	30% - 35% of Total Cost	20% - 30% of Total Cost	5% - 10% Cash Margin	35% - 50% Promoter Contribution	Unitholder Equity Capital
Indicative Benchmark Rates	**EBLR / MCLR +**	EBLR / MCLR +	EBLR / MCLR +	9.50% - 12.50%	10.00% -	**7.50% - 8.75%**

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Structural Parameter	HAM (Hybrid Annuity Model)	BOT-Toll Model	BOT-Annuity Model	Pure EPC Works Credit	Corporate Balance Sheet Term Loan	InvIT Post-COD Refinancing
	1.25% - 2.25% p.a.	2.50% - 4.00% p.a.	1.75% - 2.75% p.a.	p.a. (CC/WCDL)	13.50% p.a.	p.a. (InvIT NCD Bonds)
Construction Moratorium	Construction Period + 6 Months	Construction Period + 6 Months	Construction Period + 6 Months	None (30-90 days WCDL)	6 - 18 Months	None (Already operational COD assets)
Door-to-Door Debt Tenure	**12 - 15 Years** post-COD	15 - 20 Years post-COD	12 - 15 Years post-COD	1 - 3 Years Revolving Line	5 - 8 Years	15 - 20 Years (Bond Maturation)
Primary Collateral Security	Concession Rights + TRA Escrow + Charge on Receivables	Toll Plaza Receipts + Concession Rights + TRA Escrow	Annuity Receipts + Concession Rights + TRA Escrow	Hypothecation of WIP & Certified RA Bills	First Mortgage on Factory Land & Plant	Pledge of Project SPV Shares & Cash Flows
Target Minimum DSCR	**1.25x - 1.35x** (Base Case)	1.40x - 1.60x (High Risk Cushion)	1.20x - 1.30x	1.35x - 1.50x (ICR > 2.0x)	1.35x - 1.50x	1.30x - 1.40x
Tax Shield & Deductions	Sec 36(1)(iii) Interest + Sec 80-IA / MAT	Sec 36(1)(iii) Interest + Sec 80-IA	Sec 36(1)(iii) Interest + Sec 80-IA	Sec 36(1)(iii) Interest	Sec 36(1)(iii) Interest	Dividend / Interest Tax Exemption

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Module 4: Common Eligibility Criteria, Credit Appraisal & Underwriting Framework

4.1 Quantitative Financial Underwriting Metrics & Formulas

Credit appraisal of HAM project finance requires a rigorous cash-flow modeling framework. Lenders do not rely on net profit margins or book asset values; instead, underwriting centers on three core coverage ratios:

1. Cash Flow Available for Debt Service (CFADS)

CFADS measures the net operational cash generated by the SPV in a specific annuity period available to service senior debt:

$$extCFADS_t = extAnnuityReceipts_t + extInterestAnnuity_t + extO\&MAnnuity_t - extStatutoryTaxes_t - extActualO\&MExpenses_t - extInsuranceOutflows_t$$

2. Debt Service Coverage Ratio (DSCR)

The DSCR measures the SPV's margin of safety in meeting bi-annual debt servicing obligations (principal installment + interest payment):

$$extDSCR_t = racextCFADS_t extSeniorPrincipalRepayment_t + extSeniorInterestPayment_t$$

- **Base Case DSCR Requirement**: Lenders mandate a minimum average DSCR of **1.25x to 1.35x** over the entire 15-year door-to-door loan tenure.
- **Floor DSCR Requirement**: In any single worst-case semi-annual annuity cycle, the DSCR must not drop below **1.15x**. If projected DSCR falls below 1.15x, the sanctioned loan amount is automatically downsized until coverage compliance is achieved.

3. Loan Life Coverage Ratio (LLCR)

LLCR assesses the net present value (NPV) of all future CFADS over the remaining debt life relative to the outstanding principal balance:

$$extLLCR_t = rac\sum_{i=t}^{N} racextCFADS_i\left(1 + K_d\right)^i extOutstandingSeniorDebtBalance_t$$

Where K_d is the weighted average cost of senior debt. Lenders mandate an LLCR of $\geq 1.35x$.

4.2 Promoter Eligibility & Qualitative Credit Scorecard

While HAM debt is non-recourse to the parent company during the operational phase, the creditworthiness, technical track record, and financial strength of the sponsor group are thoroughly appraised for the construction phase:

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1. **Past Execution Track Record**: The promoter group must demonstrate completion of at least 2 to 3 major highway EPC or BOT projects of similar scale (minimum 50% to 100% of the bid lane-kilometers) within the preceding 5 fiscal years.
2. **Promoter Financial Capacity**: The parent entity must possess a Net Worth equal to at least **25% to 30% of the Total Project Cost** of all ongoing and under-bidding projects combined.
3. **Credit Bureau & CIBIL Commercial Rank**:
    - Promoter Personal CIBIL Score: ≥ 720.
    - Corporate CIBIL Commercial Rank (CMR): **CMR 1 to CMR 3** (indicating low probability of default).
4. **External Credit Rating**: The sponsor entity must carry an external investment-grade credit rating from CRISIL, ICRA, CARE, or India Ratings of **CRISIL BBB+ or higher** (preferably A- or AA category).

4.3 Pre-Disbursal Conditions Precedent (CPs) & Right of Way (ROW) Thresholds

A major cause of historical infrastructure delays was prematurely drawing bank debt before land was acquired. Under current RBI and NHAI guidelines, lenders strictly enforce mandatory Conditions Precedent (CPs) before releasing the initial loan drawdown:

- **80% Unencumbered Right of Way (ROW)**: Under NHAI HAM concession terms, the Appointed Date (official start of construction) cannot be declared, and bank debt cannot be disbursed, until NHAI hands over physical, unencumbered possession of at least **80% of the project land** (verified via 3H and 3G awards under the National Highways Act, 1956).
- **Environmental & Statutory Clearances**: Final Environment Clearance (EC) from MoEFCC, Forest Clearance (Stage I & Stage II), Wildlife Sanctuaries NOC, and Railway Overbridge (ROB) approval drawings from Indian Railways.
- **Upfront Promoter Equity Injection**: Deposit of 50% to 100% of the total required promoter cash equity into the TRA Escrow Account.
- **Executed Project Contracts**: Fully executed LSTK EPC Contract, Substitution Agreement, and TRA Escrow Agreement in place.

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Module 5: Security Architecture, Documentation & Legal Framework

5.1 Primary Security Package & Legal Mortgages

In BOT / HAM Project Funding, lenders construct a multi-layered, first-ranking security structure over all physical, financial, and contractual assets of the SPV:

1. **Section 58(f) Equitable Mortgage**: Creation of a first-ranking equitable mortgage by deposit of title deeds (under Section 58(f) of the Transfer of Property Act, 1882) over all immovable property of the SPV, including leasehold land rights, highway structures, toll plazas, and administrative buildings.
2. **First Exclusive Hypothecation Charge**: Created under Section 2(1)(n) of the SARFAESI Act, 2002, over all movable fixed assets, plant & machinery, construction equipment, unbilled Work-in-Progress (WIP), book debts, and all bank accounts (including the TRA Escrow Account, DSRA, and MMRA).
3. **Absolute Assignment of Project Contracts**: Registered assignment of all contractual rights, titles, benefits, and claims of the SPV arising under:
    - Concession Agreement executed with NHAI/MoRTH.
    - Tripartite Substitution Agreement (granting lenders the right to step in and replace the promoter upon default).
    - Lump-Sum Turnkey (LSTK) EPC Contract and Performance Guarantees.
    - Operations & Maintenance (O&M) Contracts.
    - Comprehensive Project Insurance Policies (Loss Payee Clause in favor of Lead Bank).
4. **Pledge of Equity Shares**: Pledge of **51% to 100% of the total issued equity share capital** and preference shares of the SPV held by the promoter group, executed under Sections 172 and 176 of the Indian Contract Act, 1872.
5. **Debt Service Reserve Account (DSRA)**: Lenders mandate the creation of a DSRA funded with an amount equal to **1 to 2 semi-annual principal and interest debt servicing installments** (typically 6 to 12 months of debt service). The DSRA must be created upfront at COD using project cash flows or backed by an unconditional, irrevocable Bank Guarantee (BG) from a top-tier scheduled commercial bank.

5.2 Mandatory Statutory Filings & Legal Covenants

To enforce legal perfection of security and establish absolute priority over third-party creditors, lenders mandate strict statutory compliance:

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- **ROC Form CHG-1 Registration**: Under Section 77 of the Companies Act, 2013, every charge created on the SPV's assets must be registered with the Registrar of Companies (ROC) via Form CHG-1 within **30 days of charge creation**. Failure to register invalidates the charge against the Official Liquidator in the event of winding up.
- **CERSAI Portal Registration**: Registration of security interests on the Central Registry of Securitisation Asset Reconstruction and Security Interest of India (CERSAI) portal within 30 days under Chapter IV of the SARFAESI Act, 2002.
- **National E-Governance Services Limited (NeSL) Filing**: Mandatory uploading of financial debt contracts and default thresholds onto NeSL's Information Utility (IU) platform under Section 215 of the Insolvency and Bankruptcy Code (IBC), 2016.

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Module 6: Indicative Tenure, Pricing Mechanics, Accounting & Tax Shields

6.1 Door-to-Door Tenure & Sculpted Amortization

The tenure of HAM project loans is specifically tailored to align with the 15-year operational concession period awarded by NHAI:

- **Construction Phase Moratorium**: 24 to 36 months (matching the civil execution timeline) during which principal repayment is deferred, and interest during construction (IDC) is funded via bank debt tranches or equity.
- **Post-COD Grace Period**: A 6-month buffer post-Commercial Operation Date prior to the commencement of the first principal installment.
- **Repayment Period**: **12 to 15 years** (24 to 30 semi-annual installments) structured in a **sculpted amortization schedule**. Principal installments step up gradually over time to match NHAI's Asset Cost Annuities, ensuring a stable DSCR across the entire loan life.

6.2 Pricing Benchmarks & Interest Rate Reset Mechanics

Pricing on HAM project debt reflects the transitioning risk profile between construction and operations:

- **Construction Phase Spread**: Benchmark Rate (1-Year MCLR or EBLR) + **1.50% to 2.50% p.a.** (reflecting execution and completion risk).
- **Operational Phase Spread**: Upon achieving COD, the credit rating of the SPV is upgraded (typically to **CRISIL AAA or AA** due to NHAI annuity backed cash flows), triggering an automatic rate reduction down to Benchmark Rate + **0.75% to 1.50% p.a.**
- **Reset Frequency**: Annual or bi-annual interest rate resets, aligning with NHAI's Interest Annuity rate adjustments (linked to the RBI Bank Rate).

6.3 Ind AS Accounting & Corporate Tax Shields

1. Ind AS 115 / IFRIC 12 Service Concession Accounting

Under Indian Accounting Standards (Ind AS 115 / IFRIC 12), HAM projects are accounted for under the **Financial Asset Model** rather than as tangible fixed assets. Because the SPV does not own the highway (the public highway reverts to NHAI at concession end), the project cost is recognized on the balance sheet as a **Financial Receivable** due from NHAI. Revenue is recognized over the construction phase based on the stage of completion (Percentage of Completion Method - POCM).

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2. Ind AS 23 Borrowing Cost Capitalization

Under Ind AS 23, all interest costs and processing fees incurred during the construction phase are capitalized into Capital Work-in-Progress (CWIP) / Financial Receivable, avoiding P&L charge-offs before revenue generation begins.

3. Section 36(1)(iii) Income Tax Shield

Under Section 36(1)(iii) of the Income Tax Act, 1961, all interest paid on project debt during the operational phase is **100% tax-deductible** as a business expense.

$$extEffectivePost\ -\ TaxCostofDebt\left(K_{d,extpost}\right) = K_{d,extnominal} imes\left(1\ -\ T_{extcorp}\right)$$

For a corporate tax rate of 25.168% (inclusive of surcharge and cess under Section 115BAA):

$$extIfNominalDebtRate\left(K_{d,extnominal}\right) = 9.25\% extp.a.$$

$$extEffectivePost\ -\ TaxRate\left(K_{d,extpost}\right) = 9.25\% imes(1\ -\ 0.25168) = 6.9215\% extp.a.$$

This $233 extbps$ interest tax shield significantly enhances the net return on equity (ROE) for the promoter group.

4. Section 80-IA Infrastructure Tax Deductions & MAT

Infrastructure SPVs executing HAM concessions are eligible for tax deductions under Section 80-IA of the Income Tax Act, 1961, allowing 100% tax exemption on operational profits for 10 consecutive assessment years out of 15 years. While Minimum Alternate Tax (MAT) under Section 115JB applies, accumulated MAT credits can be carried forward for 15 assessment years.

5. GST Input Tax Credit (ITC) Recovery

Under Section 16 of the Central Goods and Services Tax (CGST) Act, 2017, the SPV is entitled to **100% Input Tax Credit (ITC)** on the 18% GST charged by civil contractors on EPC construction invoices, offsetting the output GST levied on NHAI annuity receipts.

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Module 7: Suitable Sectors, Borrower Profiles & Market Alignment

7.1 Suitable Infrastructure Sub-Sectors in India

While originally pioneered for national highways, the Hybrid Annuity Model (HAM) has been expanded across multiple infrastructure asset classes under the Government of India's National Infrastructure Pipeline (NIP) and National Monetisation Pipeline (NMP):

1. **National Highways & Expressways**: Executed under NHAI and MoRTH (e.g., Delhi-Mumbai Expressway, Bengaluru-Chennai Expressway, Amritsar-Jamnagar Corridor). Represents over 75% of total HAM debt volume in India.
2. **State Highway & Economic Corridors**: Executed by State Road Development Corporations (e.g., MSRDC in Maharashtra, UPSHA in Uttar Pradesh, KRDCL in Karnataka) for state highway widening and industrial corridor bypasses.
3. **Sewage Treatment Plants (STP) & Urban Water Infrastructure**: Executed under National Mission for Clean Ganga (NMCG), Jal Jeevan Mission, and AMRUT 2.0 under the "One City One Operator" HAM model (e.g., Haridwar, Varanasi, Mathura, Kanpur STPs). Private concessionaires build and operate STPs, receiving 40% construction grants and 15-year availability-based annuities tied to water quality standards.
4. **Urban Metro Rail & Bus Rapid Transit (BRTS)**: Concessions for metro track laying, signaling, and rolling stock procurement under availability-based hybrid annuity structures.
5. **Smart City Water Supply Grids & Solid Waste Management**: 24x7 pressurized drinking water supply networks and municipal solid waste-to-energy processing plants.

7.2 Target Borrower Profiles & Market Alignment

BOT / HAM Project Funding is ideally suited for: - **Established Infrastructure EPC Contractors**: Mid-to-large civil engineering companies (e.g., HG Infra, GR Infraprojects, PNC Infratech, KCC Buildcon, Dilip Buildcon) seeking to transition from pure civil contractors into asset-owning infrastructure developers. - **Diversified Infrastructure Conglomerates**: Large corporate entities (e.g., Larsen & Toubro, Adani Road Transport, Cube Highways, IRB Infrastructure) building large-scale national transport portfolios. - **Global Infrastructure Private Equity Funds**: Institutional investors (e.g., KKR, Macquarie, Blackstone, Brookfield, Actis) acquiring operational HAM assets post-COD to bundle into Infrastructure Investment Trusts (InvITs).

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Module 8: Key Advantages, Limitations & Risk Management Matrix

8.1 Key Strategic Advantages of BOT / HAM Funding

1. Complete Elimination of Traffic Volume Risk

Unlike BOT-Toll projects where revenue fluctuates with traffic downturns, economic recessions, or fuel price shocks, HAM revenue is 100% availability-based. NHAI pays the full annuity regardless of actual traffic volume, eliminating demand risk for developers and lenders.

2. Substantially Lower Developer Equity Burden

Because NHAI provides a 40% upfront construction grant, the developer needs to arrange only 60% of the project cost. On a ₹1,000 Crore project, the developer's cash equity requirement drops from ₹250–₹300 Crores (under BOT-Toll) down to just **₹150 Crores** (15% of BPC).

3. Automatic Interest Rate & Inflation Protection

NHAI's interest annuity is contractually linked to the RBI Bank Rate / MCLR, while O&M annuities are indexed 70% to WPI and 30% to CPI. This built-in inflation and interest rate pass-through insulates the SPV's DSCR from macroeconomic volatility.

4. Sovereign / AAA Credit Quality Off-Taker

NHAI is an autonomous statutory body under MoRTH, carrying a 100% quasi-sovereign credit rating. Debt service defaults due to off-taker non-payment are virtually non-existent in NHAI-awarded HAM projects.

8.2 Limitations & Risk Management Matrix

Risk Category	Key Risk Exposure	Impact on SPV / Debt Service	Mitigation Mechanism & Institutional Safeguard
Land Acquisition / ROW Delay	Delay in handing over 80% unencumbered land by NHAI.	Prolonged pre-constructio n phase, idle site overheads.	**Mandatory CP Threshold**: Lenders do not disburse debt until NHAI legally hands over 80% ROW (3H/3G awards).

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Risk Category	Key Risk Exposure	Impact on SPV / Debt Service	Mitigation Mechanism & Institutional Safeguard
Construction Cost Overrun	Spike in steel, cement, bitumen, or labor costs during execution.	Capex budget deficit, equity dilution.	**Fixed-Price LSTK Contract**: EPC contractor executes back-to-back fixed-price contract with 10% Liquidated Damages (LD) BG.
O&M Escalation / Major Maintenance	Pavement deterioration requiring premature resurfacing (overlay).	High cash drain during Year 6 and Year 12.	**Major Maintenance Reserve Account (MMRA)**: Lenders mandate reserving cash 2 years prior to scheduled overlay cycle.
PCOD / COD Certification Lag	Delay in Independent Engineer issuing Completion Certificate.	Delay in first NHAI annuity receipt.	**6-Month Post-COD Debt Moratorium**: First bank principal installment falls due 6 months after scheduled COD.
O&M Performance Deductions	Quality deductions by NHAI due to potholes or rough riding index.	Minor reduction in bi-annual O&M annuity.	**O&M Contractor Performance BG**: Back-to-back performance guarantees provided by the O&M agency.

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Module 9: Comprehensive Real-World Case Study (Indian Rupee Models)

9.1 Project Profile & Scenario Description

To illustrate the quantitative mechanics, cash flow modeling, and financial superiority of structured HAM project finance, consider the real-world case of **Mumbai & Konkan Highway Infrastructure SPV Ltd. (MKHISL)**, a Special Purpose Vehicle promoted by a mid-sized Indian infrastructure contractor (*Konkan Infrastructure & Civil Projects Ltd.*).

MKHISL was awarded a **100-km 4-Lane Highway Expansion Project** in Maharashtra under NHAI's Hybrid Annuity Model (HAM) Concession Framework.

Key Project Parameters:

- **Concession Period**: 17 Years (2.5 Years / 30 Months Construction Phase + 14.5 Years Operational Annuity Phase).
- **Appraised Bid Project Cost (BPC)**: **₹1,000.00 Crores** (Rupees One Thousand Crores).
- **NHAI Construction Grant (40%)**: **₹400.00 Crores** disbursed in 5 equal tranches of ₹80.00 Crores tied to 20%, 40%, 60%, 80%, and 100% physical execution.
- **Net SPV Outlay Balance (60%)**: **₹600.00 Crores**.
- **Annual Quoted O&M Cost**: **₹10.00 Crores / year** (indexed to WPI/CPI).
- **NHAI Interest Annuity Benchmark**: RBI Bank Rate + 3.00% = **9.50% p.a.**

9.2 Comparative Financial Options

Option A: Corporate Balance Sheet Term Loan Structure (Un-Structured)

The promoter attempts to fund the project using a traditional parent-level corporate balance sheet term loan without creating a dedicated HAM SPV project finance facility: - **Sponsor Cash Equity Contribution (40%)**: **₹400.00 Crores** (heavy corporate cash equity lockup). - **Corporate Term Loan (60%)**: **₹600.00 Crores @ 13.00% p.a.** floating interest rate. - **Repayment Terms**: 5-Year door-to-door tenure with a brief 12-month grace period, requiring equal annual principal amortization of **₹120.00 Crores / year** starting in Year 2.

Option B: Structured Non-Recourse SPV HAM Consortium Loan Structure

The promoter executes a structured HAM SPV Project Finance Consortium Loan under NHAI guidelines: - **NHAI Construction Grant (40%)**: **₹400.00 Crores** (disbursed in 5 progress-linked tranches of ₹80.00 Crores). - **Promoter Cash Equity Margin (15% of**

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BPC / 25% of SPV Outlay): **₹150.00 Crores** (preserving ₹250.00 Crores in corporate cash liquidity!). - **Senior SPV Bank Debt (45% of BPC / 75% of SPV Outlay)**: **₹450.00 Crores** provided by a bank consortium. - **Interest Rates**: **9.25% p.a.** during construction; stepping down to **8.25% p.a.** post-COD upon rating upgrade to CRISIL AA. - **Repayment Terms**: 30-month construction moratorium (interest capitalized into CWIP under Ind AS 23) followed by **14 years (28 semi-annual installments)** of sculpted principal amortization matching NHAI's Asset Cost Annuities.

9.3 Detailed INR Cash Flow & DSCR Mathematical Model

Construction Phase (Months 1 to 30):

- Total Construction Outlay: **₹1,000.00 Crores**.
- Sourced via:
  - NHAI Construction Grant: **₹400.00 Crores**
  - Sponsor Cash Equity: **₹150.00 Crores**
  - Senior SPV Debt Drawdowns: **₹450.00 Crores**
- **Under Option B**, 100% of construction interest (₹62.40 Crores cumulative IDC) is capitalized into Capital Work-in-Progress under Ind AS 23. Zero cash debt service is required from the SPV during construction.
- **Under Option A**, the parent company must pay ₹78.00 Crores annual cash interest out of existing corporate P&L, creating severe liquidity distress before the highway generates a single rupee.

Operational Annuity Phase (Year 3 Post-COD Onwards):

- **Annual NHAI Asset Cost Annuity Receipt**:

$$extAssetAnnuity = racext₹600.00CroresBalance15extYears = ₹40.00extCrores/year$$

- **Annual NHAI Interest Annuity Receipt (Year 3 - Initial Balance ₹600 Cr @ 9.50%)**:

$$extInterestAnnuity = ₹600.00extCroresimes9.50\% = ₹57.00extCrores/year$$

- **Annual NHAI O&M Annuity Receipt (with 5% Inflation Adjustment)**:

$$extO\&MAnnuity = ₹10.00extCroresimes1.05 = ₹10.50extCrores/year$$

- **Total Gross Annual NHAI Cash Receipts (Year 3)**:

$$extTotalNHAIInflows = ₹40.00 + ₹57.00 + ₹10.50 = ₹107.50extCrores/year$$

- **Less Actual SPV O&M & Statutory Outflows**:

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$$extActualO\&MExpenses = ₹9.50extCrores$$

$$extInsurance\&Taxes = ₹6.00extCrores$$

$$extTotalOperationalOutflows = ₹15.50extCrores$$

- **Net Cash Flow Available for Debt Service (CFADS)**:

$$extCFADS = ₹107.50extCrores - ₹15.50extCrores = ₹92.00extCrores/year$$

9.4 Debt Servicing & Coverage Ratio Comparison

Option A Performance (Corporate Term Loan):

- Year 3 Principal Repayment Obligation: **₹120.00 Crores**
- Year 3 Interest Outflow (₹480 Cr Balance @ 13.00%): **₹62.40 Crores**
- Total Annual Debt Servicing Obligation: **₹182.40 Crores**
- **Year 3 DSCR (Option A)**:

$$extDSCR_{extOptionA} = racext₹92.00Croresext₹182.40Crores = 0.5044x \quad (extSEVERECASHDEFAULT\&INSOLVENCY!)$$

> *Result under Option A*: The company faces a massive **₹90.40 Crore cash deficit** in Year 3 alone, triggering loan default, credit rating downgrade to D (Default), and NCLT insolvency proceedings.

Option B Performance (Structured SPV HAM Consortium Loan):

- Senior Debt Balance at COD: **₹450.00 Crores**.
- Year 3 Principal Repayment Installment (14-Year Amortization): **₹25.00 Crores**.
- Year 3 Senior Interest Outflow (₹450 Cr @ 8.25% p.a. Post-COD Rate): **₹37.125 Crores**.
- Total Annual Senior Debt Servicing Obligation: **₹62.125 Crores**.
- **Year 3 DSCR (Option B)**:

$$extDSCR_{extOptionB} = racext₹92.00Croresext₹62.125Crores = 1.4809x \quad (extPRIMEINVESTMENTGRADE!)$$

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9.5 Wealth Surplus & Strategic Opportunity Analysis

FINANCIAL COMPARISON & WEALTH SURPLUS		
Metric / Parameter	**Option A (Corporate)**	**Option B (Structured)**
Total Initial Promoter Cash Equity Required	₹400.00 Crores	₹150.00 Crores
Upfront Cash Equity Preserved for Promoter	INR 0.00	**+₹250.00 Crores**
Year 3 Operational DSCR	**0.5044x (DEFAULT)**	**1.4809x (PRIME)**
Post-Tax Effective Cost of Debt (Sec 36(1)(iii))	9.728% p.a.	6.172% p.a.
Reinvestment Returns on Preserved Equity (@ 16%)	INR 0.00	**+₹200.00 Crores**
Direct Interest & Servicing Cash Savings	INR 0.00	**+₹99.96 Crores**
TOTAL OPPORTUNITY-ADJUSTED WEALTH SURPLUS	**BASELINE**	**+₹549.96 CRORES**

Total Financial Wealth Surplus Calculation:

$$ext{TotalNetWealthSurplus} = ext{PreservedEquity}ext{₹250.00Cr} + ext{ReinvestmentIncome}ext{₹200.00Cr} + ext{InterestSavings}ext{₹99.96Cr} = ₹549.96ext{CRORES}$$

By choosing **Option B (Structured Non-Recourse SPV HAM Consortium Loan)**, *Mumbai & Konkan Highway Infrastructure SPV Ltd.* generates a staggering **₹549.96 Crores (+INR 549,96,00,000.00) in net financial wealth surplus**, preserves ₹250.00 Crores in corporate cash, prevents a catastrophic cash default, and achieves a prime operational DSCR of 1.48x!

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Module 10: Regulatory Compliance, RBI IRACP Norms & Default Framework

10.1 Reserve Bank of India (Project Finance) Directions 2025/2026

Under the Reserve Bank of India (Project Finance) Master Directions, banks and NBFCs financing infrastructure concession projects must adhere to strict provisioning and monitoring rules:

- **Construction Phase Provisioning**: Lenders must maintain a **5.0% standard asset provision** on all outstanding project finance debt during the construction phase (up from 0.40%).
- **COD Step-Down Provisioning**: Upon successful achievement of Commercial Operation Date (COD), standard asset provisioning steps down to **2.5%**.
- **Stabilized Operational Provisioning**: Upon two years of successful operational performance with a sustained DSCR of $\geq 1.25x$, standard asset provisioning drops to the baseline **1.0%**.

10.2 Income Recognition, Asset Classification and Provisioning (IRACP) Norms

Under RBI IRACP guidelines, an infrastructure term loan or cash credit facility is classified as a **Non-Performing Asset (NPA)** if: - Principal or interest installment remains overdue for **more than 90 consecutive days**. - In the case of HAM accounts, if NHAI delays annuity payment, lenders grant a cure period matching NHAI's contractual grace period (typically 30 days) before initiating IRACP downgrades.

10.3 Extension of Date of Commencement of Commercial Operations (DCCO)

If an infrastructure project faces execution delays due to factors beyond the control of the promoter (such as court injunctions, land acquisition delays by NHAI, or force majeure events), RBI guidelines permit banks to extend the DCCO: - **Permissible Extension**: Up to **2 years** for infrastructure projects without automatically downgrading the account classification to NPA, provided the debt is restructured within RBI's framework.

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Module 11: Professional Certification Examination & Detailed Rationales

This module contains 10 scenario-based examination questions designed to test comprehensive understanding of BOT / HAM Project Funding.

Question 1

Under the Hybrid Annuity Model (HAM) framework mandated by NHAI, what percentage of the Bid Project Cost (BPC) is provided by the government authority as a construction grant during the execution phase? A) 20% B) 40% C) 60% D) 80%

Answer: B Rationale: Under NHAI's HAM framework, the government authority provides 40% of the Bid Project Cost as a construction grant disbursed in 5 equal tranches of 8% tied to physical progress milestones. The balance 60% is funded by the developer SPV.

Question 2

How does NHAI calculate the bi-annual Interest Annuity paid to the SPV during the 15-year operational phase of a HAM highway project? A) Fixed 5% flat rate on total project cost. B) RBI Bank Rate + 3.00% (or average 1-year MCLR) calculated on the reducing balance of the 60% unpaid asset cost. C) WPI inflation index minus 2.00%. D) Based on actual vehicular toll collections at the toll plaza.

Answer: B Rationale: NHAI pays interest on the reducing balance of the 60% unpaid asset cost at a rate equal to the RBI Bank Rate + 3.00% (or top 5 commercial banks' average MCLR), protecting lenders and developers against interest rate volatility.

Question 3

In a HAM project with an appraised Bid Project Cost of ₹1,000 Crores, what is the typical minimum promoter cash equity required under bank financing terms (assuming a 75:25 debt-to-equity ratio on the SPV's outlay balance)? A) ₹400 Crores B) ₹250 Crores C) ₹150 Crores D) ₹50 Crores

Answer: C Rationale: The SPV must fund 60% of the ₹1,000 Crore cost (₹600 Crores). With a 75:25 debt-to-equity ratio on the SPV outlay, equity is 25% of ₹600 Crores = ₹150 Crores (which represents exactly 15% of the Total Bid Project Cost).

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Question 4

Under Indian Accounting Standards (Ind AS 115 / IFRIC 12), how is a HAM highway concession accounted for on the SPV's balance sheet? A) Tangible Fixed Asset (Property, Plant & Equipment) depreciated over 15 years. B) Financial Asset Receivable due from NHAI. C) Intangible Asset amortized based on vehicular traffic volume. D) Inventory held for resale.

Answer: B Rationale: Because the SPV does not own the public highway and NHAI contractually guarantees annuity payments, the project is accounted for under the Financial Asset Model as a Financial Receivable from NHAI.

Question 5

What is the primary purpose of a Debt Service Reserve Account (DSRA) in a structured HAM project loan? A) To pay dividends to promoter shareholders during construction. B) To hold cash funds equal to 1 to 2 semi-annual principal and interest installments to cushion against delayed annuity payments by the off-taker. C) To fund political campaign contributions. D) To pay civil sub-contractors for material cost overruns.

Answer: B Rationale: A DSRA holds 6 to 12 months of principal and interest debt service in cash or Bank Guarantees to ensure lenders receive uninterrupted debt servicing even if NHAI annuity disbursements are delayed due to administrative processing.

Question 6

Under Section 36(1)(iii) of the Income Tax Act, 1961, what is the effective post-tax cost of debt for a HAM SPV paying a nominal interest rate of 9.25% p.a., assuming a corporate tax rate of 25.168%? A) 9.2500% p.a. B) 7.5000% p.a. C) 6.9215% p.a. D) 5.1200% p.a.

Answer: C Rationale: Effective Post-Tax Cost of Debt = $9.25\%imes(1 - 0.25168) = 9.25\%imes0.74832 = 6.9215\%extp.a.$

Question 7

Which statutory filing is mandatory within 30 days of charge creation on an SPV's assets under Section 77 of the Companies Act, 2013, to prevent the charge from becoming void against the liquidator? A) Form AOC-4 B) Form MGT-7 C) Form CHG-1 D) Form DIR-12

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Answer: C Rationale: Section 77 of the Companies Act, 2013 mandates registering every charge created on company assets with the ROC via Form CHG-1 within 30 days of creation.

Question 8

What is the minimum unencumbered Right of Way (ROW) land handover required by NHAI before the Appointed Date can be declared and bank debt disbursed? A) 50% B) 60% C) 80% D) 100%

Answer: C Rationale: Under NHAI HAM guidelines and bank underwriting terms, NHAI must deliver physical possession of at least 80% unencumbered ROW land before construction can officially start and debt can be drawn down.

Question 9

How are O&M Annuities indexed under NHAI HAM concession contracts to cushion against inflation? A) 100% linked to US Dollar CPI. B) 70% Wholesale Price Index (WPI) + 30% Consumer Price Index (CPI). C) Fixed 3% annual escalation. D) Linked to BSE Sensex index movements.

Answer: B Rationale: NHAI O&M annuities are dynamically indexed using a composite index of 70% WPI and 30% CPI published by the RBI.

Question 10

Under the Reserve Bank of India (Project Finance) Directions 2025/2026, what is the standard asset provisioning requirement for commercial banks during the construction phase of an infrastructure project? A) 0.40% B) 1.00% C) 2.50% D) 5.00%

Answer: D Rationale: RBI guidelines mandate a 5.0% standard asset provision on outstanding project debt during the construction phase, which steps down to 2.5% at COD and 1.0% upon two years of operational stability.

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